

Cardiorenal Syndromes

Kevin Damman | John J.V. McMurray

CHAPTER OUTLINE

INTRODUCTION, 1380	ALBUMINURIA, 1387
TERMINOLOGY OF CARDIORENAL SYNDROME, 1381	TUBULOINTERSTITIAL DAMAGE IN HEART FAILURE, 1387
EPIDEMIOLOGY AND PROGNOSIS, 1381	TREATMENT OF HEART FAILURE IN PATIENTS WITH CONCOMITANT RENAL DYSFUNCTION, 1388
PATHOPHYSIOLOGY OF CONCOMITANT HEART FAILURE AND CHRONIC KIDNEY DISEASE, 1382	
WORSENING RENAL FUNCTION AND CHANGE IN RENAL FUNCTION, 1385	

KEY POINTS

- Cardiorenal syndrome represents the clinical combination of concomitant reduced heart and renal function.
- At least five subtypes of cardiorenal syndrome have been proposed.
- Worsening renal function in heart failure is not the same as acute kidney injury (AKI) in heart failure.
- Worsening renal function in the context of initiation or up-titration of renin-angiotensin-aldosterone system inhibitors in heart failure should not immediately prompt early discontinuation because the prognostic benefit is (often) maintained but requires close monitoring of kidney function.
- Pathophysiology of albuminuria in heart failure is different from that in primary kidney disease because intraglomerular pressure is presumed to be low in heart failure.

INTRODUCTION

Heart failure (HF) is a syndrome characterized by increased cardiac filling pressures, reduced cardiac output, signs and symptoms of congestion, and evidence of structural cardiac disease.^{1,2} The incidence and prevalence of HF is on the rise because of improved survival of patients with cardiovascular disease (e.g., myocardial infarction, hypertension), increased awareness, and early recognition of the syndrome, with sophisticated advanced diagnostic tests and better treatment of diagnosed HF. As such, the number of patients with HF is estimated to increase substantially in the next 10 years.³ This dramatic increase in patient numbers will pose a challenge to health care professionals, as well as to costs in the health care system, because HF care is already responsible for 1.4% of the total medical expenditures.⁴ The most important reason for these extraordinary costs are the relatively long and often repetitive admissions for HF when

patients present with acute (decompensated) HF. In the syndrome of HF, the kidney plays a crucial role. It is responsible for the maladaptive and excessive salt and water retention that feeds the vicious circle of increased filling pressures, low cardiac output, and reduced organ (and renal) perfusion, which triggers even more salt and water retention.⁵

This also indicates that from a renal perspective, kidney failure itself can also induce or accelerate the progression or development of HF, as is often the case in hemodialysis patients. However, even in the early stages of chronic kidney disease (CKD), the risk of incident HF is increased, suggesting that shared cardiovascular risk factors and/or the decrease in the glomerular filtration rate (GFR) itself may predispose to HF development.

This interplay between heart and kidney is key to the pathophysiologic processes that occur in HF and end-stage renal disease (ESRD), and the synergistic failure of both organs together has been termed “cardiorenal syndrome” (CRS).⁵⁻⁷ This chapter discusses the epidemiology, pathophysiology, and

important clinical considerations in patients with cardiorenal failure.

TERMINOLOGY OF CARDIORENAL SYNDROME

Followed by in-depth invasive hemodynamic research in the early 20th century,⁸⁻¹⁰ specific interest in the importance of cardiorenal interplay was revived at the end of the 20th century. Following a number of publications around the turn of the millennium showing that a decreased creatinine clearance or GFR was associated with worse outcome in patients with chronic HF, a consensus statement on the CRS was published.¹¹⁻¹⁴ This landmark paper, by Ronco and colleagues,¹⁴ has since led to numerous follow-up manuscripts; the current consensus is that in CRS, five subtypes exist (Table 40.1). It is important to note here that these classifications, including the term “cardiorenal syndrome,” have been proposed by experts in the field. This does not mean that in daily clinical practice, distinguishing between the exact types of CRS will help guide and choose treatment. However, it does serve as a starting point to determine which organ dysfunction is the more likely initiating trigger and which one is a consequence of the other. The subclassification is based on the time relationship between the failure of both organs, one occurring before the other, and differentiating between acute and chronic situations.

CRS type 5 includes patients who do not fit the description of the other 4 and are subject to a systemic process that involves (among other organs) heart and kidney together.¹⁵ From a biologic standpoint, one should realize that patients are not fixed within one particular type of CRS and can shift from one type to another, or even have two types at the same time. For example, a chronic HF patient who develops acute renal failure during admission with decompensated HF but who has had declining renal function over a longer period of time, preadmission, would fit both CRS types 1 and 2.^{16,17} If this acute renal failure is also associated with acute worsening of HF, this means that part of the pathophysiology also fits CRS type 3.¹⁸ This highlights that it is inherently difficult to subclassify patients within one specific subtype, let alone base treatment on this classification. For clinical practice, it would suffice to characterize a patient as having concomitant HF and renal dysfunction and to determine the initiating trigger, which also would be the target for therapy.

Since the introduction of CRS as an entity, different researchers have argued that the original definition could be broadened to include anemia also. The proposed term would then become “cardiorenal anemia syndrome” (CRAS).¹⁹ Anemia is frequent in both HF and CKD and is associated with worse outcomes.²⁰ The pathophysiology is not entirely understood and is a matter of debate, but reduced erythropoietin production and responsiveness, bone marrow suppression, and fluid retention (hemodilution) are the most prominent factors.²¹ It was also thought that iron deficiency might be important because it is frequently observed together with anemia in HF, but it is now clear that nonanemic HF patients are often iron-deficient as well.²² Therefore, some argue that the terminology could even be changed to cardiorenal anemia iron deficiency syndrome (CRAIDS) or CRIDS when anemia is absent. The more subdivisions that are proposed, the more difficult it is to define a homogeneous population in which specific treatment algorithms may be proposed. It does, however, emphasize the complex pathophysiologic relationships among HF, kidney dysfunction, and other organ dysfunction and systemic abnormalities, which probably is a consequence of the heterogeneity of the cardiorenal population.

EPIDEMIOLOGY AND PROGNOSIS

In patients with HF, renal dysfunction, mostly defined as CKD (estimated GFR [eGFR] < 60 mL/min/1.73 m²) is frequently present. In the MAGGIC meta-analysis, including unselected patient cohorts with mostly chronic HF, around 50% of patients had CKD.²³ In a larger meta-analysis, including over 1,000,000 patients with HF, CKD was present in 42% of patients with chronic HF, whereas CKD was present in 53% of patients with acute HF.²⁴ These populations consist of different phenotypes of HF, including patients with preserved ejection fraction (HF with preserved ejection fraction [HFPEF]) and patients with HF with reduced ejection fraction (HFREF; Table 40.2). Data on the comparison between these two phenotypes are scarce, but it is thought that in general, the prevalence of CKD is similar, although patient characteristics of these patient cohorts differ substantially.²³

In patients with chronic HF, when CKD is present, the risk of death is more than twofold greater, a risk estimate that is similar in patients with acute HF.²⁴ Additionally, from retrospective analyses from clinical trials, the magnitude of

Table 40.1 Classification of Cardiorenal Syndromes

Type	Popular Term	Description
1	Acute cardiorenal syndrome	Acute heart failure leading to an acute renal event, including acute kidney injury (AKI), worsening renal function
2	Chronic cardiorenal syndrome	Chronic heart failure leading to renal injury or progression of renal failure
3	Acute renocardiac syndrome	AKI leading to an acute cardiac event, including acute heart failure
4	Chronic renocardiac syndrome	Chronic kidney disease (CKD) leading to cardiac injury or progression of cardiac disease, including heart failure
5	Systemic cardiorenal syndrome	A systemic event or process that results in the simultaneous development of acute heart and kidney injury, including heart failure and AKI

Table 40.2 Definition of Phenotypes of Heart Failure

Type of Heart Failure	HFREF	HFPEF
Criteria		
1	Symptoms ± signs	Symptoms ± signs ^a
2	LVEF < 40%	LVEF ≥ 50%
3	—	Elevated levels of natriuretic peptides and either relevant structural heart disease (left ventricular hypertrophy and/or large left atrium) or diastolic dysfunction

HFREF, Heart failure with reduced ejection fraction; HFPEF, heart failure with preserved ejection fraction; LVEF, left ventricular ejection fraction.

Adapted from Ponikowski P, Voors AA, Anker SD, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure: the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC), developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur J Heart Fail*. 2016;18:891–975.

^aIn that guideline, a middle group, termed “heart failure with midrange ejection fraction” (HFmrEF), is specified. However, at present, there is little evidence that such a subgroup actually exists or affects treatment.

the greater risk for all-cause mortality (and HF hospitalization) was directly related to the degree of renal impairment at baseline, showing a stepwise increase in unadjusted hazard ratio—for example, 3.83 (2.92–5.01) for eGFR below 45 mL/min/1.73 m² compared with an eGFR > 90 mL/min/1.73 m².¹³ In a different analysis, chronic HF patients with eGFR < 15 mL/min/1.73 m² had the highest risk for cardiovascular mortality, with the risk steeply increasing when the eGFR was lower than 60 mL/min/1.73 m².²⁵

In patients with CKD, data on the prevalence of HF are scarcer. However, it is clear that with decreasing eGFR, the risk for cardiovascular disease in general increases steeply in the general population, as well as in patients. On a population level, the prevalence of HF in subjects with eGFR < 60 mL/min/1.73 m² is between 2- and 20-fold higher compared with those with preserved eGFR.²⁶ This higher prevalence of HF in individuals with a lower eGFR is not restricted to HF, because similar patterns are found for coronary artery disease, cerebrovascular events, and peripheral artery disease, although the association is less striking.

In the Chronic Renal Insufficiency Cohort (CRIC), which included almost 4000 patients with impaired eGFR, the risk of HF rose steeply with decreasing eGFR levels.²⁷ The hazard ratio (HR) for the development of HF was 1.67 (1.49–1.89) per standard deviation (15 mL/min/1.73 m²) decrease in eGFR, an association that was independent of albuminuria. In the Atherosclerosis Risk in Communities (ARIC) study, including over 15,000 patients (45–64 years of age), the risk of incident HF was more than threefold higher (HR, 3.57 [1.89–6.77]) when eGFR was < 30 mL/min/1.73 m² but no albuminuria was present, whereas the HR increased to 7.98 (3.46–18.41) when macroalbuminuria was also present.²⁸ There are even fewer data on the risk of HF when acute kidney injury (AKI) develops. In a meta-analysis of mostly individuals with cardiovascular disease, where different definitions were pooled, the relative risk for HF associated with AKI was 58% higher than in individuals with cardiovascular disease with no AKI.²⁹

PATHOPHYSIOLOGY OF CONCOMITANT HEART FAILURE AND CHRONIC KIDNEY DISEASE

SHARED RISK FACTORS

To understand the pathophysiology of the interplay between HF and renal dysfunction, once either one has developed, it is important to realize that the risk factors for both are remarkably similar. In particular, cardiovascular risk factors, such as hypertension, diabetes, peripheral artery disease, and obesity, predispose to failure of both organs. In many cases, these risk factors likely increase the risk of an acute event (e.g., myocardial infarction), which subsequently may lead to HF and/or CKD; on the other hand, chronic exposure to any of these risk factors may accelerate the slow progression to end-stage organ dysfunction (e.g., hypertension leading to left ventricular hypertrophy, hypertensive nephropathy, and HFPEF). Importantly, when a CRS has developed, these risk factors remain associated with worse outcomes.

RENAL HEMODYNAMICS

RENAL BLOOD FLOW

In patients with chronic HF (HFREF), cardiac output is markedly reduced, leading to a decrease in organ perfusion, including perfusion of the brain, gut, liver and, most importantly, the kidney.³⁰ Whereas the renal perfusion normally is around 25% of cardiac output, this proportion decreases significantly when cardiac output falls.³¹ Despite a fall in cardiac output, the kidney is able to preserve the GFR in a low-pressure, low-flow state because of renal autoregulation, although most patients are not severely hypotensive in the early stages of HF. In patients without renin-angiotensin-aldosterone system (RAAS) inhibition therapy, when renal blood flow (RBF) decreases and intraglomerular pressures fall, efferent autoregulation will increase the filtration fraction (FF), and the GFR will remain preserved to some extent and for some time (see Chapter 3).³¹ With further deterioration

in clinical status and renal perfusion (pressure), more stringent efferent and now also afferent arteriolar autoregulation will be unable to preserve GFR, and the FF will decrease. This results in loss of the GFR. Especially in these very low-flow states, where hypotension is a real issue, GFR becomes highly flow-dependent.

Clinical Relevance

In patients with heart failure who are renin-angiotensin-system (RAAS) inhibitor-naïve, a reduction in cardiac output and consequently renal blood flow (RBF) does not directly translate into a reduction in glomerular filtration rate (GFR). Because of efferent arteriolar vasoconstriction, GFR is relatively preserved, resulting in increased filtration fraction. However, in patients using RAAS inhibitor therapy, this adaptive process is blocked, resulting in a reduction in GFR in parallel to the reduction in RBF, with a similar filtration fraction. This is the reason for a reduction in GFR in (some) patients with heart failure who experience an increase in the serum creatinine level. This small decrease in GFR should be distinguished from the more clinically relevant situation of an unprovoked deterioration in GFR, which requires adjustment of treatment and more elaborate assessment of the patient's clinical status.

Things are slightly different in patients who are treated with RAAS inhibitors. Angiotensin-converting enzyme (ACE) inhibitors and angiotensin II type 1 receptor blockers (ARBs) inhibit the efferent arteriolar vasoconstriction. This means that in the situation of reduced RBF, the expected efferent arteriolar vasoconstriction is prevented, resulting in loss of GFR (Fig. 40.1).³² Only when RBF drops significantly further will the situation be similar to that without RAAS inhibitors;

both efferent and afferent vasoconstriction progress (possibly due to tubuloglomerular feedback) and cause further decreases in GFR, FF, and very low intraglomerular pressures.⁷ It is under these circumstances, in particular, that chronic HF patients may develop albuminuria.³² It must be emphasized that this situation is opposite from the albuminuria that is present in hypertension, diabetes, and CKD, where high intraglomerular pressures exist; in HF, in contrast, low intraglomerular pressures are present. The actual cause of albuminuria in HF is therefore not entirely understood but could include glomerular (endothelial) damage and leakage, possibly influenced by the low-grade systemic inflammation present in HF, and abnormal tubular handling of filtered albumin or tubular dysfunction.

There have been no studies investigating renal hemodynamics in HF patients with preserved ejection fraction. It is expected that especially in the end stage of HFPEF, renal hemodynamics behave similarly to HFREF patients, because cardiac output is likely significantly reduced in these patients as well. Whether RBF is reduced in mild HFPEF is unknown. Similarly, in acute HF, where assessment of renal hemodynamics is difficult, no studies have investigated renal perfusion and GFR simultaneously. However, at least 50% of patients presenting with acute HF are actually patients with acute exacerbation of chronic HF. In such patients, RBF will already be reduced chronically. Among patients presenting with new-onset acute HF, the extent of the reduction in RBF is unknown.

In patients with CKD-AKI who develop HF (CRS types 3 and 4)^{18,33} or CRS type 5,¹⁵ no specific studies have been conducted on renal perfusion and association with GFR. CRS types 3 and 4 are discussed further in Chapter 54.

CENTRAL AND RENAL VENOUS PRESSURE

In addition to a reduced cardiac output, patients with HF often experience not only increased left-sided filling pressure,

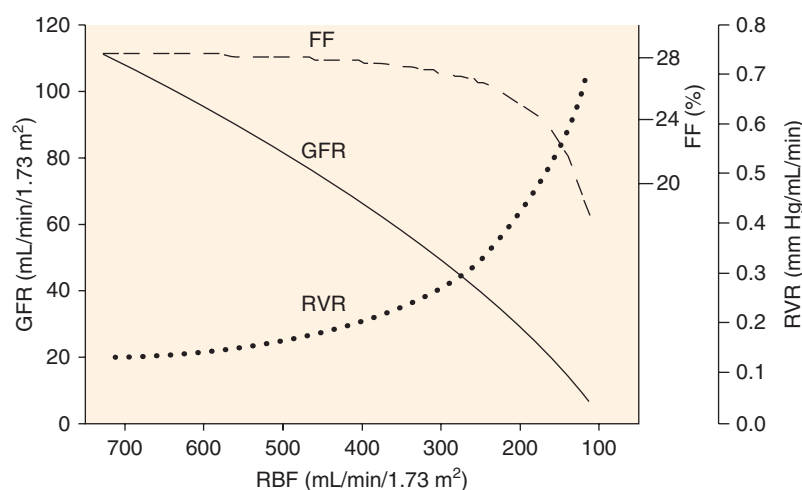


Fig. 40.1 Renal hemodynamics in chronic HFREF in the presence of RAAS inhibition. Shown is the relationship among RBF, RVR, FF, and GFR in the presence of RAAS inhibition. With **low RBF**, GFR decreases although FF is maintained when efferent arteriolar vasoconstriction is inhibited. Only when RBF decreases substantially will the FF decrease because of **increased afferent arteriolar vasoconstriction**. This results in increased RVR, especially in those patients who experience albuminuria. FF, Filtration fraction; GFR, glomerular filtration rate; RAAS, renin-angiotensin-aldosterone system; RBF, renal blood flow; RVR, renal vascular resistance. (From Smilde TD, Damman K, van der Harst P, et al. Differential associations between renal function and “modifiable” risk factors in patients with chronic heart failure. *Clin Res Cardiol.* 2009;98:121–129. Reproduced with permission from Springer.)

but also increased central venous pressure (CVP). This is the case for chronic HF patients who are relatively stable, but is even more important in patients presenting with acute or worsening HF, in whom increased CVP is one of the hallmarks of this syndrome. Because the kidney is dependent on a pressure gradient across the glomerular membrane to allow passive filtration, any change in pressure inside the glomerulus or pressure in the Bowman capsule may influence filtration. Because the renal perfusion pressure is dependent on the differences in mean arterial pressure and renal venous pressure and renal vascular resistance, a change in RBF may not only induce changes in GFR, but also change in central and renal venous pressures. This concept has long been overlooked in modern medicine. Already, in the early part of the 20th century, extensive measurements were carried out (in patients without RAAS inhibitors), and the association between higher CVP and higher renal venous pressure was described.^{8-10,34,35} Even more importantly, a higher renal venous pressure was associated with lower RBF and, subsequently, lower GFR. Therefore, because of a reduction in perfusion pressure across the kidney due to higher CVP, perfusion is decreased, and GFR deteriorates. This is the RBF-dependent effect of an increased renal venous pressure.

However, there is also an RBF-independent effect of increased renal venous pressure on the GFR. An increase in renal venous pressure leads to an increase in renal interstitial pressure.^{36,37} Because the kidney is an encapsulated organ, this will lead to an increase in the pressure in the Bowman capsule, as well as lead to collapse of tubules that are compressed by the higher interstitial pressures. We now also appreciate that in patients with HF, the pressure wave caused by severe tricuspid regurgitation, together with central venous flow pattern alterations occurring in individuals with high CVP, actually reach the renal parenchyma.^{38,39} These flow patterns in the large caval veins alter with a higher CVP and have been associated with worse outcomes in HF patients. Although uncertain, it is possible that this will further compromise renal function.

High central and renal venous pressures are now considered important pathophysiologic factors in patients with HF and renal dysfunction.^{16,17,40} In two simultaneously published studies, higher CVP was an important predictor of worsening renal function in acute HF, whereas CVP was also the strongest independent factor associated with eGFR in a large heterogeneous cardiovascular population in another study.^{41,42} There is even evidence that increased renal venous pressure, by itself, is associated with salt and water retention.⁴³ Although the importance of a higher CVP in relation to lower eGFR is especially important in the acute situation, where CVP is increased, it must be acknowledged that a reduction in RBF is still probably the most important determinant of GFR in any patient with HF.⁴⁴ Higher CVP does not predispose to worse renal function in every patient, because the opposite has also been found in different studies.⁴⁵ Future studies are needed to establish whether changes in CVP (reduction in the acute HF situation) are associated with changes in GFR, although some data do support this.^{46,47} There have been no studies directly measuring renal venous pressure or renal interstitial pressure in HF or CRS. Fig. 40.2 summarizes the pathophysiology for CRS types 1 and 2 in patients with HF. For CRS types 3 and 4, where the first event is acute or chronic renal failure or injury, it is unknown

whether compromised renal hemodynamics play a role in the pathophysiology of CRS. AKI certainly can predispose to salt and water retention, thereby increasing CVP and cardiac filling pressures. For CRS type 5, systemic processes themselves, such as shock, changes in microcirculation, redistribution of fluid to the splanchnic system resulting in hypotension, or treatment of these processes, including vasopressor therapy and fluid resuscitation, may cause higher CVP, lower renal blood flow, and induce cardiac injury.¹⁵ However, the relative contribution and importance of these processes have not been established and therefore are speculative, at best.

CARDIORENAL CONNECTORS

In patients with HF, it is likely that the hemodynamic variables, RBF and increased CVP, are the most important factors determining GFR. There are, however, systemic modulating factors that have been designated as “cardiorenal connectors.”⁵ These include endothelial dysfunction, inflammation, sympathetic nervous system (SNS) activation, RAAS activation, and reactive oxygen species (ROS) activation. These cardiorenal connectors do not only directly affect GFR, but also affect RBF, thereby also influencing GFR indirectly. In CRS types 3 and 4, levels of different inflammatory markers and markers of endothelial dysfunction are increased, and there is experimental evidence suggesting that some of these proteins (e.g., tumor necrosis factor- α , interleukin-6) have detrimental effects on cardiac function.^{18,33}

Other data have suggested that uremic CKD patients experience cardiac (and renal) fibrosis, which is at least partly attributable to upregulation of the transforming growth factor-beta (TGF- β) pathway.^{33,48} Inflammation by itself can also trigger more pronounced RAAS activation and the formation of ROS. RAAS activation is not only characterized by systemic effects of angiotensin II, but also by intracardiac and intrarenal effects. Angiotensin II induces cardiac remodeling, modulates the effect of renal interstitial hypertension,³⁷ and triggers vascular inflammation through the nuclear factor kappa B (NF- κ B) pathway.^{5,49}

Oxidative stress has been shown to influence cardiac function negatively and is associated with cardiac mitochondrial dysfunction.⁵⁰ Furthermore, not only is systemic ROS formation greatly increased in experimental models of CRS, but intrarenal oxidative stress also occurs, perhaps reflecting increased levels of nicotinamide adenine dinucleotide phosphate (NADPH) oxidase activity.⁵⁰ Sustained activation of the SNS is prevalent in HF and CRS and is the target of therapies such as beta-blockers. SNS activation has detrimental effects on cardiac and renal function as a consequence of recurrent cardiac and renal ischemia and mediation of fibrosis, together with RAAS activation and oxidative stress.^{5,51} Furthermore, SNS activation can modify the ultrafiltration coefficient and lead to salt and water retention.⁵² The detailed biologic processes and interplays between the different cardiorenal connectors are probably highly complex and are certainly poorly understood, as are their relative importance in the heart and renal impairment in CRS. It seems, however, probable that these cardiorenal connectors, together with the shared risk factors, modulate the hemodynamic relationship between heart and kidney in CRS and that activation of one system may result in activation of several others.

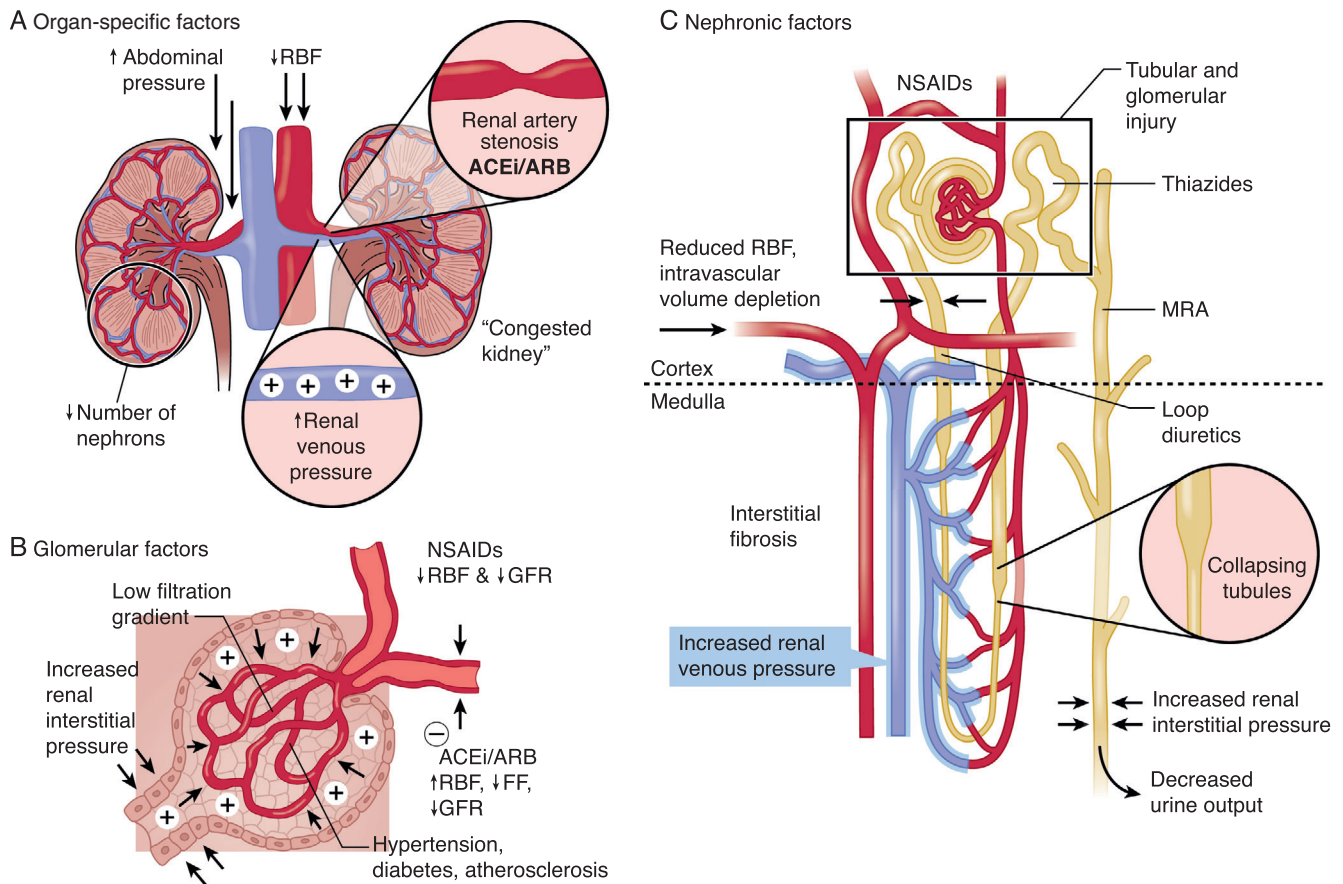


Fig. 40.2 Factors involved in the cause of and association with outcomes of changes in renal function in heart failure. (A) Organ-specific factors. The main determinants of decreased glomerular filtration rate are a decrease in renal blood flow and an increase in central and renal venous pressure. The latter can be caused by intravascular congestion but also by an increase in intraabdominal pressure. Owing to increased renal venous pressure, renal interstitial pressure rises, which results in a so-called congested kidney, because the kidney is encapsulated (B and C). Renal artery stenosis is present in around 25% of heart failure patients, which can further compromise renal blood flow, especially in the presence of renin-angiotensin-aldosterone system (RAAS) inhibitors. (B) Glomerular factors. Decreased renal blood flow and low blood pressure trigger renal autoregulation, preserving the glomerular filtration rate by increasing the filtration fraction by increased efferent vasoconstriction. The use of RAAS inhibitors inhibits this process, which increases renal blood flow, but leads (in some patients) to a reduction in glomerular filtration rate (pseudo-worsening renal function). Nonsteroidal antiinflammatory drugs inhibit prostaglandin synthesis, thereby impairing prostaglandin-associated increased or dependent renal blood flow. Increased interstitial pressure causes increased pressure in the Bowman capsule, which directly opposes filtration and in a glomerulus, where the filtration gradient is already low due to a decreased renal blood flow and increased renal venous pressure. Concomitant diseases have direct but differential effects on glomerular filtration, glomerular integrity, and podocyte function, as well as autoregulation. (C) Nephronic factors. Different therapies have different renal effects and exert their action at specific sites, as indicated in this diagram. Intravascular volume depletion (in the presence or absence of congestion) can lead to impaired renal perfusion and decreased glomerular filtration rate. The combination of increased interstitial pressure, reduced arterial perfusion, and concomitant disease, and therapies can cause tubular and glomerular injury. Increased renal venous pressure causes increased renal interstitial pressure, resulting in collapsing of renal tubules, which decreases the glomerular filtration rate and eventually leads to decreased urine output, sodium retention, and congestion. ACEi, Angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; FF, filtration fraction; GFR, glomerular filtration rate; MRA, mineralocorticoid receptor antagonist; NSAIDs, nonsteroidal antiinflammatory drugs; RAAS, renin-angiotensin-aldosterone system; RBF, renal blood flow. (From Damman K, Tang WH, Testani JM, et al. Terminology and definition of changes renal function in heart failure. *Eur Heart J*. 2014;35:3413–3416. Reprinted with permission from Oxford University Press.)

WORSENING RENAL FUNCTION AND CHANGE IN RENAL FUNCTION

Not only is CKD prevalent in patients with HF, but accelerated decline in eGFR is frequently observed in these patients. There are few data from unselected patient cohorts showing overall change in eGFR across the HF population, but it has been estimated that the decrease in eGFR/year could be as high as 2 mL/min/1.73 m².²⁵ However, how fast eGFR may

decrease depends on the clinical setting. For example, in patients admitted for acute HF, the change from admission to follow-up after 18 months in surviving patients was −7 mL/min/1.73 m², of which −2 mL/min/1.73 m² occurred in the hospital.⁵³ In retrospective analyses of large clinical trials, mostly with RAAS inhibitor therapy, long-term changes in eGFR have been investigated. For example, in the CHARM program, which included both HFPEF and HFREF patients, the mean change in eGFR/year was −2.8 mL/min/1.73 m², with a more significant deterioration in eGFR with the ARB

candesartan compared with placebo.⁵⁴ The retrospective analysis from the GISSI-HF study in over 6000 HF patients had the benefit that the randomized treatment did not directly affect the slope of eGFR and, in this particular trial, the mean change per year in eGFR was $-2.6 \text{ mL/min/1.73 m}^2$.²⁵ Only a few patients (15%) had a large decrease in eGFR in the first year ($>15 \text{ mL/min/1.73 m}^2$ decrease), and 11% of patients actually had an increase in eGFR of the same magnitude. The greater the decrease in eGFR after 12-month follow-up, the worse the clinical outcome. Factors related to a steeper decline in eGFR in this particular analysis were the presence of obstructive lung disease, more preserved renal function at baseline, and the use of loop diuretics.

How much eGFR changes in the hospital in HF patients is difficult to establish. The serum creatinine level can fluctuate greatly in these patients during the initial phase of admission and treatment, and it is a matter of debate whether these changes in creatinine level really reflect changes in intrinsic glomerular filtration function. What we do know from large observational studies in acute HF is that the serum creatinine level may show different patterns of change, depending on patient characteristics, choice of treatment, hemodynamic alterations, and comorbid organ dysfunction.

It has, therefore, been difficult to establish the normal change in creatinine level and eGFR in patients with acute HF and how different patterns relate to differential outcomes. A significant increase in the serum creatinine level is observed in around 20% to 25% of all patients with HF.²⁴ Definitions vary, but mostly include an absolute increase of $>0.3 \text{ mg/dL}$ ($26.5 \mu\text{mol/L}$) and/or a relative increase in the serum creatinine level and eGFR over a certain period of time. In patients with HF, this has been termed “worsening renal function,” which is distinctly different from AKI that occurs in CRS type 3 or primary kidney disease.⁵⁵ There are similarities, but the main difference seems to be that in AKI, there is evidence of actual kidney injury, with oliguria and anuria, histopathologic evidence of renal damage, and a very short time course. For worsening renal function, there is much more evidence for a transient functional change in GFR that is reversible most of the time.

The proposed definition to be used in HF differs slightly from that of AKI definitions in the nephrology literature.⁵⁵ Most important, there are additional criteria stating that a deterioration in creatinine level should be accompanied by a deterioration in clinical status to be designated as worsening renal function. Multiple studies and meta-analyses have evaluated worsening renal function in acute and chronic HF and found that overall worsening renal function is associated with worse outcomes.^{24,56} However, it is now becoming clear that if the HF status of a patient improves despite the occurrence of worsening renal function, worsening renal function itself is not associated with worse outcomes. For example, in patients with acute HF, in whom the serum creatinine level can increase for various reasons, including intravascular volume depletion, loop diuretic therapy, and changes in hemodynamics, worsening renal function in these settings is not associated with increased mortality if the response to loop diuretics (decongestion) is adequate.^{57,58}

Similarly, the serum creatinine level often increases after beginning and uptitration of RAAS inhibitor therapy. In contrast to patients in whom worsening renal function is unprovoked, the patients with RAAS inhibitor-induced worsening

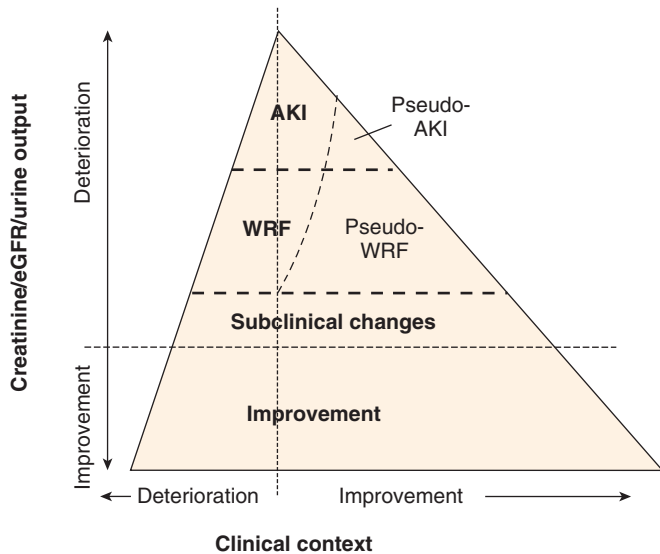


Fig. 40.3 Visual depiction of association among changes in renal function, clinical condition, and mortality risk. Only when both deterioration in clinical status and increase in the serum creatinine level (or decrease in renal function) track together is this associated with worse clinical outcomes in heart failure. AKI, Acute kidney injury; GFR, glomerular filtration rate; WRF, worsening renal function. Darker colors indicate higher mortality risk. (From Damman K, Testani JM. The kidney in heart failure: an update. *Eur Heart J*. 2015;36:1437–1444. Reprinted with permission from Oxford University Press.)

renal function have favorable outcomes.⁵⁹ The reason for the apparently paradoxical association between deterioration in eGFR with improvement in outcomes during treatment with these agents is the attenuation of the autoregulatory response to reduced RBF, as discussed earlier. This provoked worsening of renal function without a deterioration in clinical status has now been termed “pseudo-worsening renal function” (Fig. 40.3). It highlights the importance of not relying solely on serum creatinine measurements, but also incorporating information on the patient’s clinical status when evaluating the relevance of any change in renal function.⁵⁵ It must, however, be emphasized that RAAS inhibitor-induced worsening renal function can only be termed “pseudo-worsening renal function” in HFREF patients when the clinical outcome is improved with these agents; in patients with HFPEF, the worsening renal function observed with ARBs was actually associated with worse outcome.⁶⁰ Whenever the serum creatinine level does increase after initiation or uptitration of RAAS inhibitor therapy, it is important to monitor the serum creatinine and electrolyte levels (e.g., potassium) and clinical status of these patients closely.^{2,61,62} An increase as much as 50% above baseline serum creatinine, or up to 3 mg/dL ($266 \mu\text{mol/L}$) and $\text{eGFR} < 25 \text{ mL/min/1.73 m}^2$ is acceptable if the clinical status requires no adjustment. Changes in renal function that exceed these thresholds should be regarded as clinically significant and are a reason to lower the dose of RAAS inhibitor therapy. If the serum creatinine level increases by more than 100% or exceeds 3.5 mg/dL ($310 \mu\text{mol/L}$), and $\text{eGFR} < 20 \text{ mL/min/1.73 m}^2$, RAAS inhibitor therapy needs to be stopped temporarily. Renal function and electrolytes should be checked on a regular basis thereafter and, if possible, patients should be rechallenged

to evaluate whether the intolerance to RAAS inhibitors was temporary or permanent. In the situation of excessive or unexpectedly large increases in the serum creatinine level, clinicians should always consider alternative reasons, such as interactions with other medications, renal artery stenosis, dehydration, or decompensation and adherence issues. These factors should be investigated and possibly treated before rechallenge is started.

DETERIORATION OF CARDIAC FUNCTION

In CKD, ESRD, or dialysis, there is direct evidence that maladaptive processes that normally occur during normal aging, or in hypertension, diabetes, and atherosclerosis, are markedly accelerated. Whether this is also the case in AKI is incompletely understood. In patients with ESRD or dialysis, left ventricular hypertrophy (LVH) is not only frequently observed, but there is also evidence that the progression to (more severe) LVH is often seen when eGFR decreases or (hemo)dialysis is initiated.⁶³ LVH is a consequence of myocyte hypertrophy, which results in myocardial stiffening and is associated with myocardial fibrosis and (subendocardial) ischemia due to myocyte–capillary mismatch. In CKD and ESRD, myocyte hypertrophy and myocardial fibrosis occur due to RAAS activation and upregulation of profibrotic pathways, which result in higher circulating levels of profibrotic molecules, such as TGF- β , galectin-3, and angiotensin II.¹⁸ Over two-thirds of ESRD patients exhibit LVH, of which 30% have manifest LV failure. In one study of almost 600 patients, LV mass increased by over 60% 1.5 years after the initiation of dialysis, whereas almost 50% of patients exhibited evidence of HF.⁶⁴ On the other hand, among 160 patients from CRIC, there was no difference in LV mass between CKD and ESRD.⁶⁵ However, LVEF did decrease marginally from 53% to 50% in the transition to ESRD. More sophisticated echocardiographic assessments, including global longitudinal strain (GLS) measurements, have shown that with more severe renal disease, overall cardiac contractility deteriorates, even in patients with preserved LVEF.⁶⁶ Finally, as discussed above, AKI and CKD predispose to more frequent HF events (the hallmark of CRS types 3 and 4), suggesting a temporal relationship between the progression of renal disease and deterioration in cardiac function.

ALBUMINURIA

Albuminuria and proteinuria frequently develop and are hallmarks of CKD. As such, in patients who have AKI or CKD complicated by HF (CRS type 3 or 4), albuminuria is often present and has been the focus of extensive research. In patients with HF complicated by CKD, worsening renal function, or AKI, albuminuria has been observed as well.^{67,68} In patients with chronic HFREF or HFPEF, three retrospective analyses from randomized studies have shown that albuminuria is present in up to 41% of patients (20%–30% microalbuminuria, 5%–11% macroalbuminuria).^{67,69} Similar prevalences were observed in one single-center study of stable HFPEF patients.⁷⁰ In acute HF, fewer data are available. Using a dipstick test to evaluate proteinuria, 55% of patients with acute HF had a positive dipstick test in a Chinese cohort.⁷¹ In a small cohort of 100 acute HF patients from Portugal,

only 26% to 40% of patients with acute HF presented with normoalbuminuria.⁷² In all these analyses, including any HF phenotype, the presence of albuminuria was associated with a higher risk of cardiovascular events, independent of baseline eGFR. Furthermore, neither ARB nor statin treatment was able to reduce the urinary albumin excretion rate significantly.^{67,68}

The pathophysiology of increased albumin excretion in HF, as discussed earlier, is probably different from albuminuria in hypertension, diabetes, and other cardiovascular disease, including CKD without HF. Whereas in these conditions it is thought that higher intraglomerular pressures (either primarily or as an adaptive process) induce structural glomerular changes, podocyte dysfunction, and eventually leakage of macromolecules such as albumin, in HF the intraglomerular pressures are (probably) low. The reasons for low intraglomerular pressures are a low RBF, a low efferent arteriolar tone in the presence of RAAS inhibitor therapy, and probably high afferent arteriolar tone in very low cardiac output states. Indeed, in the most severe chronic HFREF patients, RBF, GFR, and FF are low, whereas renal vascular resistance and albumin excretion are high.³² This suggests that the pathophysiology of albumin excretion in HF must be distinctly different from that observed in hypertension, diabetes, and intrinsic renal disease. Proposed mechanisms include endothelial dysfunction and inflammation, structural glomerular damage, and changes in the glycocalyx, as well as podocyte dysfunction.

Whether higher renal venous pressure and associated higher renal interstitial pressures also contribute to albuminuria is unknown. Most interestingly, and reinforcing the notion that albuminuria must be different in HF compared with CKD, sacubitril/valsartan treatment in HFPEF and HFREF actually increases urinary albumin excretion, but leads to substantially reduced rates of adverse cardiovascular outcomes.^{73,74} This essentially indicates that together with a lack of effect of both statin and ARB treatment on albumin levels, albuminuria by itself serves as a risk marker, rather than as a risk factor, in HF.⁷⁵

TUBULOINTERSTITIAL DAMAGE IN HEART FAILURE

Analogous to CKD, patients with HF experience not only a reduced eGFR and albuminuria, but also exhibit evidence of tubulointerstitial damage and injury.⁷⁶ However, in contrast to primary kidney disease, histologic evidence of tubulointerstitial damage is rarely obtained because biopsies are seldom performed in patients with HF. In one small study in patients with severely reduced eGFR considered for heart transplantation, renal biopsies were performed.⁷⁷ The cause of HF differed, ranging from ischemic HF to dilated cardiomyopathy and amyloid disease. The renal pathology did not correlate directly with cardiac pathology. Renal pathology ranged from ischemic nephropathy to focal global glomerulosclerosis and diabetic glomerulosclerosis. Fibrosis in the renal biopsies of the patients with severe HF ranged from 5% to 50%, but did not correlate with the amount of proteinuria or degree of eGFR decrease. These data indicate that in patients with severe HF, renal fibrosis is often present but does not correlate with a reduction in eGFR and albuminuria, unlike in primary

CKD, reflecting the superimposed physiologic component of renal dysfunction.

Most indirect evidence of tubulointerstitial damage and injury in HF patients comes from biomarker analyses. In parallel to studies in CKD, changes in urinary (and plasma) proteins that have been associated with tubular injury and damage in patients with kidney disease can also be found in a large proportion of patients with HF. In chronic HFREF patients, urinary levels of markers of tubulointerstitial damage, such as kidney injury molecule 1 (KIM-1), neutrophil gelatinase-associated lipocalin (NGAL), and N-acetyl-beta-D-glucosaminidase (NAG), were markedly elevated compared with normal values from control subjects.^{76,78-81} Notably, there was only a marginal correlation between these markers of tubulointerstitial damage and albuminuria.⁸² All measures of renal dysfunction, including reduced eGFR, albuminuria, and increased levels of tubulointerstitial damage, provided prognostic information in these patients. In the longer term, higher urinary KIM-1 levels were associated with more frequent occurrence of worsening renal function in the chronic outpatient setting, even outperforming baseline eGFR in predicting this.⁷⁶

In acute HF, evidence of tubulointerstitial damage is clear as well.^{78,79,83} When present either at presentation or at discharge, evidence of tubular injury predisposes to worse outcomes in these patients, especially higher mortality.^{78,84} In addition to providing prognostic information, these markers of tubulointerstitial damage, especially NGAL, have been shown to predict AKI in different patient settings without HF, including patients undergoing thoracic surgery and patients with primary kidney disease.^{85,86} However, the prediction of AKI and worsening renal function in acute HF has been more challenging, and these markers have not been shown to be of any help in predicting clinically relevant worsening renal function. Although small studies have suggested possible incremental information obtained with determining NGAL levels, two large studies have now shown lack of superiority of NGAL over the serum creatinine level. In the prospective AKINESIS study, plasma NGAL was not superior to the serum creatinine level in predicting AKI in those with acute HF, whereas in a retrospective analysis from the PROTECT study, serial plasma NGAL concentrations rose in parallel with the serum creatinine level in patients with worsening renal function, but did not outperform the serum creatinine level.^{87,88} One small study has evaluated changes in these markers in response to loop diuretic withdrawal and re-initiation in chronic HF and showed that both NAG and KIM-1 showed significant changes in response to changes in diuretic treatment, whereas the serum creatinine level did not change.⁸⁹ Whether tubulointerstitial damage may be a target for therapy has not been established in HF.

TREATMENT OF HEART FAILURE IN PATIENTS WITH CONCOMITANT RENAL DYSFUNCTION

In patients with acute HF who develop worsening renal function (CRS type 1), treatment is not different from that given to acute HF patients without renal dysfunction. Because no evidence-based treatments that improve outcome have been established in acute HF (rather than chronic HF), therapy is

aimed at improvement of symptoms. It is, however, important to realize that patients with renal impairment at presentation (or that develops during admission) may require different doses of diuretics, vasodilators, and other therapies because their response to these treatments may be different. Most of the time these treatments will be less effective. For example, diuretic resistance and a reduced response to loop diuretic therapy in these patients is observed much more frequently. Furthermore, as alluded to in the previous sections, an increase in the serum creatinine level during decongestive therapy, without a deterioration in clinical status (or actual improvement), is often observed and acceptable. Often, these changes are transient and, when these occur in parallel, with a favorable diuretic response, are not associated with worse outcomes. It has not been established which dosage (high vs. low) or which administration type (bolus or continuous) of loop diuretics is superior, although higher dosages were associated with more frequent worsening renal function, without translating into a difference in clinical outcome.⁹⁰

In the situation of chronic HF patients with renal dysfunction (CRS type 2), a distinction must be made for the different phenotypes of chronic HF. Treatment for HFPEF patients, with or without renal dysfunction, should focus on the underlying disease and treatment of comorbidities because there are no evidence-based treatments that improve outcomes in these patients.¹

The cornerstone treatments in all patients with HFREF include RAAS inhibitors such as ACE inhibitors, ARBs, or preferably the combination of a neprilysin inhibitor with an ARB (i.e., sacubitril/valsartan), as well as beta-blockers and mineralocorticoid receptor antagonists (Fig. 40.4 and Table 40.3).^{1,2,62} Most recent HF guidelines advocate the start of ACE inhibitor therapy (or ARB if ACE inhibitor-intolerant), together with beta-blocker therapy. These agents can safely be prescribed in HF patients with eGFR > 30 mL/min/1.73 m², and there is moderate to strong evidence of mortality benefit in HF patients with CKD stages 1 to 3. Combination therapy of ACE inhibition and ARB therapy is not recommended and is associated with a higher risk of worsening renal function and hyperkalemia, especially in CRS type 2.⁶¹ When these treatments are at their maximum tolerable dose, and patients remain symptomatic, the next step will be addition of a mineralocorticoid receptor antagonist. This means that on top of ACE inhibition, ARB, and beta-blocker therapy, either spironolactone or eplerenone is added. In several studies in HF patients with CKD stages 1 to 3, these treatments were effective in reducing the occurrence of (cardiovascular) death or HF rehospitalization.⁹¹ The risk of hyperkalemia with this drug combination, however, must be borne in mind, especially at lower GFRs. If despite these agents, HFREF patients remain symptomatic, the next step would be to replace ACE inhibitor or ARB therapy with sacubitril/valsartan, which is a combination of an ARB (valsartan) with a neprilysin inhibitor. This combination treatment was shown to reduce the risk of (cardiovascular) death as compared with enalapril in patients with CKD stages 1 to 3 and reduced the risk of renal adverse events.⁷⁴

For patients who have both HFREF and reduced eGFR (CRS type 2), these treatments will generally be (and should be) prescribed without hesitation. However, none of the evidence-based treatments in HFREF have been evaluated in HF patients with eGFR below 30 mL/min/1.73 m², which

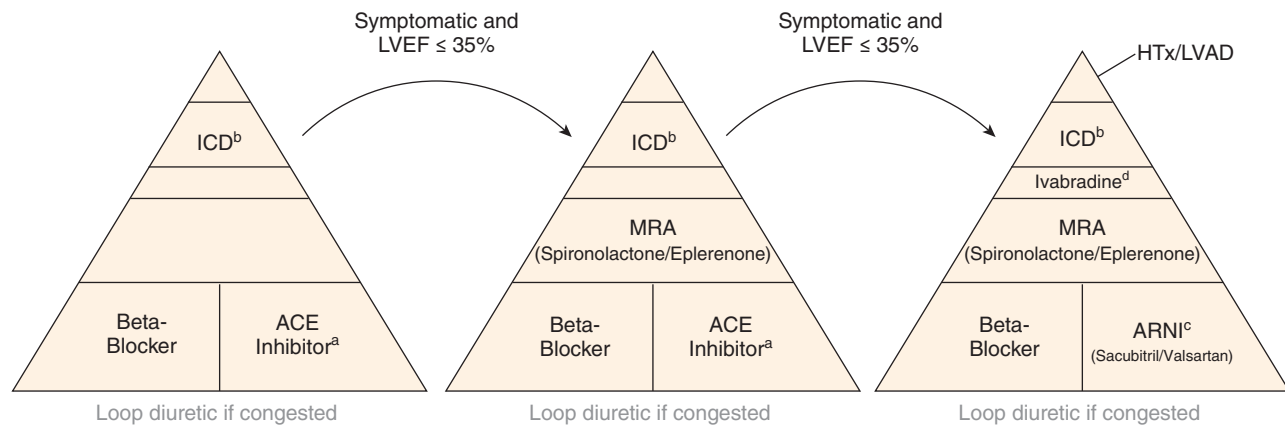


Fig. 40.4 Treatment algorithm for HFREF patients. This is the treatment algorithm for HFREF patients in general and for patients with CKD up to stage 3 (stages 4 and 5 not included). Initial treatment should include an ACE inhibitor and beta-blocker, uptitrated to the maximal tolerated dose. Loop diuretics are advised to relieve symptoms and signs of congestion. ACE, Angiotensin-converting enzyme; CKD, chronic kidney disease; HFREF, heart failure with reduced ejection fraction; HTx, heart transplantation. ^aSwitch or start with an ARB if the ACE inhibitor is not tolerated. Do not combine an ACE inhibitor and ARB. ^bICD (implantable cardioverter-defibrillator) indicated in secondary prevention or in symptomatic patients with LVEF ≤ 35%, despite optimal medical treatment (cardiac resynchronization therapy in selected patients). If patients remain symptomatic, add a mineralocorticoid receptor antagonist (MRA) and uptitrate to the maximum tolerated dose. ^cIf, despite these therapies, the patient remains symptomatic, the ACE inhibitor can be replaced by ARNI (angiotensin receptor blocker neprilysin inhibitor—sacubitril/valsartan). ^dIn patients with sinus rhythm and heart rate >70 beats/min, ivabradine may be added. Only select patients with advanced refractory heart failure, LVAD, and/or heart transplantation may be considered. (Adapted from Ponikowski P, Voors AA, Anker SD, Bueno H, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure: the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC). Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur J Heart Fail.* 2016;18:891–975. Also adapted from references 2, 61, and 62.)

means there is no solid scientific evidence for the use of these agents in HF patients with stage 4 or 5 CKD.⁹¹ Furthermore, clinicians may be reluctant to maintain patients on RAAS inhibitor therapy when the serum creatinine level increases during uptitration or to start these agents when eGFR is close to 30 mL/min/1.73 m². However, the latter group represents patients who will probably benefit even more from evidence-based therapies compared with those with relatively preserved eGFR, and therefore could be initiated at low doses and uptitrated, with close follow-up.

Other therapies, such as implantable cardioverter-defibrillators (ICDs) and/or cardiac resynchronization therapy (CRT), should not be withheld in patients with concomitant cardiac and renal failure if life expectancy is sufficient to allow benefit in quality of life or life expectancy.⁹¹ Some treatments, such as CRT and left ventricular assist devices (LVADs), have actually been shown to improve renal function in the long term.⁹² LVADs are essentially extracardiac pumps that restore (part of) the circulation and thereby organ perfusion. Even though they do not provide an artificial pulse, but rather a continuous flow, renal function is often restored (to some extent), although AKI and/or worsening renal function is also often observed in the early postoperative phase of the implantation of the device. The finding that renal function improves over time with these devices suggests that for CRS type 2, impaired hemodynamics caused by worse LV function result in renal dysfunction by causing worse renal perfusion.

For patients with advanced HF who are considered possible heart transplant recipients, a greatly reduced GFR is an important relative contraindication to transplantation. Immunosuppression after transplantation is often nephrotoxic and, together with the cardiac transplant procedure itself,

may initiate or exacerbate renal injury and dysfunction. A low eGFR in patients with HF is often due to hemodynamic derangement and can be restored after heart transplantation, but other comorbid organ dysfunctions, such as diabetes, atherosclerosis, and hypertension, may also have caused irreversible renal damage. The current criteria for heart transplant listing suggest thorough evaluation of renal function, and only list patients who do not have irreversible stage 4 or 5 CKD.⁹³ In select patients, usually young patients, a combined heart and kidney transplant may be considered, but this decision has to be made on a case-by-case basis, also taking into account donor organ availability, quality, and recipient characteristics.

For HF patients with severe CKD, treatment with evidence-based therapies should always be an individualized decision based on the balance between the risk of worsening heart and renal function and possible benefit expected with each treatment. This may mean that in the situation of palliative care in patients with very severe HF and/or renal disease, some evidence-based treatments that preserve renal and cardiac function and are associated with better outcomes in the long term may be discontinued because of worsening renal function, hypotension, and/or other discomforts to improve quality of life.

In conclusion, CRSs represent a very heterogeneous group of disorders in patients with both heart and kidney dysfunction. This chapter specifically focused on CRS types 1 and 2, where heart–kidney interactions are predominantly mediated by impaired hemodynamics. Although treatment of HF patients with renal dysfunction is in itself no different from treatment of patients without CKD, there is limited evidence of the effectiveness of these treatments in CKD stages 4 and 5. Because HF patients with concomitant renal

Table 40.3 Characteristics of Different Treatment in HFREF Patients With Chronic Kidney Disease

Therapy	Incidence of Worsening Renal Function and Adverse Events	Incidence of Hyperkalemia	Effectiveness in HFREF Patients ^a		Cautions and Remarks
			CKD Stage 1–3	CKD Stage 4 or 5	
ACE inhibitor	1.5%–13.7% (35% in NYHA IV)	1.1%–6.4% (7% in NYHA IV)	Yes	Unclear; possible	Induces early decline in eGFR; some increase in serum creatinine should be accepted. Very large increases should prompt further investigation and (temporary) stopping of drug.
ARB	5.5%–17% (24% with high-dose losartan)	1%–3% (10% with high-dose losartan)	Yes	Unclear	
MRA	1.9%–17%	2%–8%	Yes	Unclear; possible	
ARNI	2.2%	4.3% (potassium > 6 mmol/L)	Yes	Unclear; possible	
Beta-blocker	7%–10.1%	NA	Yes	Probable	Effect on renal function negligible compared with placebo; should be continued if possible.
Loop diuretics	NA	Probably low	NA	NA	Use and dose associated with worsening renal function. Long-term effects on renal function unknown. Dose should be higher in patients with CKD stage 3–5.
CRT	NA	NA	Yes	Unclear; possible	Improvement in renal function in parallel—improvement in clinical symptoms can be expected.
LVAD	NA	NA	Yes	Unclear; possible	LVAD therapy improves renal function in the long term. However, risk of AKI is peri- and postoperatively higher in patients with CKD stage 3–5 at baseline. Risk of contrast nephropathy at time of implantation.

^aImprovement in clinical outcome.

ARB, Angiotensin II receptor blocker; ACE, angiotensin-converting enzyme; ARNI, angiotensin receptor blocker neprilysin inhibitor; CKD, chronic kidney disease; CRT, cardiac resynchronization therapy; eGFR, estimated glomerular filtration rate; HFREF, heart failure with reduced ejection fraction; LVAD, left ventricular assist device; NYHA, New York Heart Association.

Adapted from Damman K, Tang WH, Felker GM, et al. Current evidence on treatment of patients with chronic systolic heart failure and renal insufficiency: practical considerations from published data. *J Am Coll Cardiol*. 2014;63:853–871.

dysfunction represent a very high-risk population, they should be monitored and evaluated closely and on a regular basis, and treatment alterations should be based on a case-by-case evaluation.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

- Ponikowski P, Voors AA, Anker SD, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2016;37:2129–2200.
- Writing Committee Members, Yancy CW, Jessup M, et al. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on practice guidelines. *Circulation*. 2013;128:e240–e327.
- Ronco C, Haapio M, House AA, et al. Cardiorenal syndrome. *J Am Coll Cardiol*. 2008;52:1527–1539.
- Damman K, Voors AA, Navis G, et al. The cardiorenal syndrome in heart failure. *Prog Cardiovasc Dis*. 2011;54:144–153.
- Maxwell MH, Breed ES, Schwartz IL. Renal venous pressure in chronic congestive heart failure. *J Clin Invest*. 1950;29:342–348.
- Werko L, Varnauskas E, Ek J, et al. Studies on the renal circulation and renal function in mitral valvular disease. II. Effect of apresoline. *Circulation*. 1954;9:700–705.
- Blake WD, Wegria R, Keating RP, et al. Effect of increased renal venous pressure on renal function. *Am J Physiol*. 1949;157:1–13.
- Haase M, Muller C, Damman K, et al. Pathogenesis of cardiorenal syndrome type 1 in acute decompensated heart failure: workgroup statements from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:99–116.
- Cruz DN, Schmidt-Ott KM, Vescovo G, et al. Pathophysiology of cardiorenal syndrome type 2 in stable chronic heart failure: workgroup

- statements from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:117–136.
23. McAlister FA, Ezekowitz J, Tarantini L, et al. Renal dysfunction in patients with heart failure with preserved versus reduced ejection fraction: impact of the new Chronic Kidney Disease-Epidemiology Collaboration Group formula. *Circ Heart Fail*. 2012;5:309–314.
 25. Damman K, Masson S, Lucci D, et al. Progression of renal impairment and chronic kidney disease in chronic heart failure: an analysis from GISSI-HF. *J Card Fail*. 2017;23:2–9.
 26. Go AS, Chertow GM, Fan D, et al. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med*. 2004;351:1296–1305.
 30. Leithe ME, Margorien RD, Hermiller JB, et al. Relationship between central hemodynamics and regional blood-flow in normal subjects and in patients with congestive heart-failure. *Circulation*. 1984;69:57–64.
 31. Ljungman S, Laragh JH, Cody RJ. Role of the kidney in congestive heart failure. Relationship of cardiac index to kidney function. *Drugs*. 1990;39(suppl 4):10–21.
 32. Smilde TD, Damman K, van der Harst P, et al. Differential associations between renal function and “modifiable” risk factors in patients with chronic heart failure. *Clin Res Cardiol*. 2009;98:121–129.
 36. Burnett JC Jr, Knox FG. Renal interstitial pressure and sodium excretion during renal vein constriction. *Am J Physiol*. 1980;238:F279–F282.
 40. Damman K, Testani JM. The kidney in heart failure: an update. *Eur Heart J*. 2015;36:1437–1444.
 41. Mullens W, Abrahams Z, Francis GS, et al. Importance of venous congestion for worsening of renal function in advanced decompensated heart failure. *J Am Coll Cardiol*. 2009;53:589–596.
 42. Damman K, van Deursen V, Navis G, et al. Increased central venous pressure is associated with impaired renal function and mortality in a broad spectrum of patients with cardiovascular disease. *J Am Coll Cardiol*. 2009;53:582–588.
 47. Mullens W, Abrahams Z, Francis GS, et al. Prompt reduction in intra-abdominal pressure following large-volume mechanical fluid removal improves renal insufficiency in refractory decompensated heart failure. *J Card Fail*. 2008;14:508–514.
 51. McCullough PA, Kellum JA, Haase M, et al. Pathophysiology of the cardiorenal syndromes: executive summary from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:82–98.
 55. Damman K, Tang WH, Testani JM, et al. Terminology and definition of changes renal function in heart failure. *Eur Heart J*. 2014;35:3413–3416.
 56. Damman K, Navis G, Voors AA, et al. Worsening renal function and prognosis in heart failure: systematic review and meta-analysis. *J Card Fail*. 2007;13:599–608.
 61. Ponikowski P, Voors AA, Anker SD, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure: the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC). Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur J Heart Fail*. 2016;18:891–975.
 67. Masson S, Latini R, Milani V, et al. Prevalence and prognostic value of elevated urinary albumin excretion in patients with chronic heart failure: data from the GISSI-heart failure trial. *Circ Heart Fail*. 2010;3:65–72.
 68. Jackson CE, Solomon SD, Gerstein HC, et al. Albuminuria in chronic heart failure: prevalence and prognostic importance. *Lancet*. 2009;374:543–550.
 70. Katz DH, Burns JA, Aguilar FG, et al. Albuminuria is independently associated with cardiac remodeling, abnormal right and left ventricular function, and worse outcomes in heart failure with preserved ejection fraction. *JACC Heart Fail*. 2014;2:586–596.
 74. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371:993–1004.
 82. Damman K, Masson S, Hillege HL, et al. Clinical outcome of renal tubular damage in chronic heart failure. *Eur Heart J*. 2011;32:2705–2712.
 87. Maisel AS, Wettersten N, van Veldhuisen DJ, et al. Neutrophil gelatinase-associated lipocalin for acute kidney injury during acute heart failure hospitalizations: the AKINESIS study. *J Am Coll Cardiol*. 2016;68:1420–1431.
 90. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364:797–805.
 91. Damman K, Tang WH, Felker GM, et al. Current evidence on treatment of patients with chronic systolic heart failure and renal insufficiency: practical considerations from published data. *J Am Coll Cardiol*. 2014;63:853–871.

REFERENCES

- Ponikowski P, Voors AA, Anker SD, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure: the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC) developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur Heart J*. 2016;37:2129–2200.
- Writing Committee Members, Yancy CW, Jessup M, et al. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on practice guidelines. *Circulation*. 2013;128:e240–e327.
- Heidenreich PA, Albert NM, Allen LA, et al. Forecasting the impact of heart failure in the United States: a policy statement from the American Heart Association. *Circ Heart Fail*. 2013;6:606–619.
- Cook C, Cole G, Asaria P, et al. The annual global economic burden of heart failure. *Int J Cardiol*. 2014;171:368–376.
- Bongartz LG, Cramer MJ, Doeveandans PA, et al. The severe cardio-renal syndrome: 'Guyton revisited.' *Eur Heart J*. 2005;26:11–17.
- Ronco C, Haapio M, House AA, et al. Cardiorenal syndrome. *J Am Coll Cardiol*. 2008;52:1527–1539.
- Damman K, Voors AA, Navis G, et al. The cardiorenal syndrome in heart failure. *Prog Cardiovasc Dis*. 2011;54:144–153.
- Maxwell MH, Breed ES, Schwartz IL. Renal venous pressure in chronic congestive heart failure. *J Clin Invest*. 1950;29:342–348.
- Werko L, Varnauskas E, Ek J, et al. Studies on the renal circulation and renal function in mitral valvular disease. II. Effect of apresoline. *Circulation*. 1954;9:700–705.
- Blake WD, Wegria R, Keating RP, et al. Effect of increased renal venous pressure on renal function. *Am J Physiol*. 1949;157:1–13.
- Dries DL, Exner DV, Domanski MJ, et al. The prognostic implications of renal insufficiency in asymptomatic and symptomatic patients with left ventricular systolic dysfunction. *J Am Coll Cardiol*. 2000;35:681–689.
- Hillege HL, Girbes AR, de Kam PJ, et al. Renal function, neurohormonal activation, and survival in patients with chronic heart failure. *Circulation*. 2000;102:203–210.
- Hillege HL, Nitsch D, Pfeffer MA, et al. Renal function as a predictor of outcome in a broad spectrum of patients with heart failure. *Circulation*. 2006;113:671–678.
- Ronco C, McCullough P, Anker SD, et al. Cardio-renal syndromes: report from the consensus conference of the Acute Dialysis Quality Initiative. *Eur Heart J*. 2010;31:703–711.
- Mehta RL, Rabb H, Shaw AD, et al. Cardiorenal syndrome type 5: clinical presentation, pathophysiology and management strategies from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:174–194.
- Haase M, Muller C, Damman K, et al. Pathogenesis of cardiorenal syndrome type 1 in acute decompensated heart failure: workgroup statements from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:99–116.
- Cruz DN, Schmidt-Ott KM, Vescovo G, et al. Pathophysiology of cardiorenal syndrome type 2 in stable chronic heart failure: workgroup statements from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:117–136.
- Bagshaw SM, Hoste EA, Braam B, et al. Cardiorenal syndrome type 3: pathophysiologic and epidemiologic considerations. *Contrib Nephrol*. 2013;182:137–157.
- Otake Y, Watanabe T, Takahashi H, et al. Association of renal tubular damage with cardio-renal anemia syndrome in patients with heart failure. *Int J Cardiol*. 2014;173:222–228.
- Groenveld HF, Januzzi JL, Damman K, et al. Anemia and mortality in heart failure patients—a systematic review and meta-analysis. *J Am Coll Cardiol*. 2008;52:818–827.
- Westenbrink BD, Visser FW, Voors AA, et al. Anaemia in chronic heart failure is not only related to impaired renal perfusion and blunted erythropoietin production, but to fluid retention as well. *Eur Heart J*. 2007;28:166–171.
- Klip IT, Jankowska EA, Enjuanes C, et al. The additive burden of iron deficiency in the cardiorenal-anemia axis: scope of a problem and its consequences. *Eur J Heart Fail*. 2014;16:655–662.
- McAlister FA, Ezekowitz J, Tarantini L, et al. Renal dysfunction in patients with heart failure with preserved versus reduced ejection fraction: impact of the new chronic kidney disease-epidemiology collaboration group formula. *Circ Heart Fail*. 2012;5:309–314.
- Damman K, Valente MA, Voors AA, et al. Renal impairment, worsening renal function, and outcome in patients with heart failure: an updated meta-analysis. *Eur Heart J*. 2014;35:455–469.
- Damman K, Masson S, Lucci D, et al. Progression of renal impairment and chronic kidney disease in chronic heart failure: an analysis from GISSI-HF. *J Card Fail*. 2017;23:2–9.
- Go AS, Chertow GM, Fan D, et al. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med*. 2004;351:1296–1305.
- He J, Shlipak M, Anderson A, et al. Risk factors for heart failure in patients with chronic kidney disease: the CRIC (Chronic Renal Insufficiency Cohort) study. *J Am Heart Assoc*. 2017;6:doi:10.1161/JAHA.116.005336.
- Waheed S, Matsushita K, Sang Y, et al. Combined association of albuminuria and cystatin C-based estimated GFR with mortality, coronary heart disease, and heart failure outcomes: the Atherosclerosis Risk in Communities (ARIC) study. *Am J Kidney Dis*. 2012;60:207–216.
- Odutayo A, Wong CX, Farkouh M, et al. AKI and long-term risk for cardiovascular events and mortality. *J Am Soc Nephrol*. 2017;28:377–387.
- Leithe ME, Margorien RD, Hermiller JB, et al. Relationship between central hemodynamics and regional blood-flow in normal subjects and in patients with congestive heart-failure. *Circulation*. 1984;69:57–64.
- Ljungman S, Laragh JH, Cody RJ. Role of the kidney in congestive heart failure. relationship of cardiac index to kidney function. *Drugs*. 1990;39(suppl 4):10–21.
- Smilde TD, Damman K, van der Harst P, et al. Differential associations between renal function and “modifiable” risk factors in patients with chronic heart failure. *Clin Res Cardiol*. 2009;98:121–129.
- Tumlin JA, Costanzo MR, Chawla LS, et al. Cardiorenal syndrome type 4: insights on clinical presentation and pathophysiology from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:158–173.
- Werko L, Varnauskas E, Eliasch H, et al. Studies on the renal circulation and renal function in mitral valvular disease. I. Effect of exercise. *Circulation*. 1954;9:687–699.
- Kinoshita M. Studies on cardiac output to blood volume, and renal circulation in chronic congestive heart failure. *Jpn Circ J*. 1968;32:249–270.
- Burnett JC Jr, Knox FG. Renal interstitial pressure and sodium excretion during renal vein constriction. *Am J Physiol*. 1980;238:F279–F282.
- Fiksen-Olsen MJ, Strick DM, Hawley H, et al. Renal effects of angiotensin II inhibition during increases in renal venous pressure. *Hypertension*. 1992;19:137–141.
- Iida N, Seo Y, Sai S, et al. Clinical implications of intrarenal hemodynamic evaluation by Doppler ultrasonography in heart failure. *JACC Heart Fail*. 2016;4:674–682.
- Nijst P, Martens P, Dupont M, et al. Intrarenal flow alterations during transition from euolemia to intravascular volume expansion in heart failure patients. *JACC Heart Fail*. 2017.
- Damman K, Testani JM. The kidney in heart failure: an update. *Eur Heart J*. 2015;36:1437–1444.
- Mullens W, Abrahams Z, Francis GS, et al. Importance of venous congestion for worsening of renal function in advanced decompensated heart failure. *J Am Coll Cardiol*. 2009;53:589–596.
- Damman K, van Deursen V, Navis G, et al. Increased central venous pressure is associated with impaired renal function and mortality in a broad spectrum of patients with cardiovascular disease. *J Am Coll Cardiol*. 2009;53:582–588.
- Firth JD, Raine AE, Ledingham JG. Raised venous pressure: a direct cause of renal sodium retention in oedema? *Lancet*. 1988;1:1033–1035.
- Damman K, Navis G, Smilde TD, et al. Decreased cardiac output, venous congestion and the association with renal impairment in patients with cardiac dysfunction. *Eur J Heart Fail*. 2007;9:872–878.
- Uthoff H, Breidhardt T, Klima T, et al. Central venous pressure and impaired renal function in patients with acute heart failure. *Eur J Heart Fail*. 2011;13:432–439.
- Nohria A, Hasselblad V, Stebbins A, et al. Cardiorenal interactions: insights from the ESCAPE trial. *J Am Coll Cardiol*. 2008;51:1268–1274.
- Mullens W, Abrahams Z, Francis GS, et al. Prompt reduction in intra-abdominal pressure following large-volume mechanical fluid removal improves renal insufficiency in refractory decompensated heart failure. *J Card Fail*. 2008;14:508–514.
- Mall G, Huther W, Schneider J, et al. Diffuse intermyocardial fibrosis in uraemic patients. *Nephrol Dial Transplant*. 1990;5:39–44.
- Rubattu S, Mennuni S, Testa M, et al. Pathogenesis of chronic cardiorenal syndrome: is there a role for oxidative stress? *Int J Mol Sci*. 2013;14:23011–23032.

50. Giam B, Kaye DM, Rajapakse NW. Role of renal oxidative stress in the pathogenesis of the cardiorenal syndrome. *Heart Lung Circ*. 2016;25:874–880.
51. McCullough PA, Kellum JA, Haase M, et al. Pathophysiology of the cardiorenal syndromes: executive summary from the eleventh consensus conference of the Acute Dialysis Quality Initiative (ADQI). *Contrib Nephrol*. 2013;182:82–98.
52. DiBona GF. Nervous kidney. Interaction between renal sympathetic nerves and the renin-angiotensin system in the control of renal function. *Hypertension*. 2000;36:1083–1088.
53. Damman K, Jaarsma T, Voors AA, et al. Both in- and out-hospital worsening of renal function predict outcome in patients with heart failure: results from the coordinating study evaluating outcome of advising and counseling in heart failure (COACH). *Eur J Heart Fail*. 2009;11:847–854.
54. Damman K, Solomon SD, Pfeffer MA, et al. Worsening renal function and outcome in heart failure patients with reduced and preserved ejection fraction and the impact of angiotensin receptor blocker treatment. *Eur J Heart Fail*. 2016;18:1508–1517.
55. Damman K, Tang WH, Testani JM, et al. Terminology and definition of changes renal function in heart failure. *Eur Heart J*. 2014;35:3413–3416.
56. Damman K, Navis G, Voors AA, et al. Worsening renal function and prognosis in heart failure: systematic review and meta-analysis. *J Card Fail*. 2007;13:599–608.
57. Testani JM, Brisco MA, Chen J, et al. Timing of hemoconcentration during treatment of acute decompensated heart failure and subsequent survival: importance of sustained decongestion. *J Am Coll Cardiol*. 2013;62:516–524.
58. Metra M, Nodari S, Parrinello G, et al. Worsening renal function in patients hospitalised for acute heart failure: clinical implications and prognostic significance. *Eur J Heart Fail*. 2008;10:188–195.
59. Clark H, Krum H, Hopper I. Worsening renal function during renin-angiotensin-aldosterone system inhibitor initiation and long-term outcomes in patients with left ventricular systolic dysfunction. *Eur J Heart Fail*. 2014;16:41–48.
60. Beldhuis I, Streng K, Ter Maaten JM, et al. Renin angiotensin system inhibition, worsening renal function and outcome in heart failure patients with reduced and preserved ejection fraction. *Circ Heart Fail*. 2017;10(2): doi:10.1161/CIRCHEARTFAILURE.116.003588.
61. Ponikowski P, Voors AA, Anker SD, et al. 2016 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure: the Task Force for the diagnosis and treatment of acute and chronic heart failure of the European Society of Cardiology (ESC). Developed with the special contribution of the Heart Failure Association (HFA) of the ESC. *Eur J Heart Fail*. 2016;18:891–975.
62. Yancy CW, Jessup M, Bozkurt B, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association task force on clinical practice guidelines and the Heart Failure Society of America. *J Card Fail*. 2017;23:628–651.
63. McGregor E, Jardine AG, Murray LS, et al. Pre-operative echocardiographic abnormalities and adverse outcome following renal transplantation. *Nephrol Dial Transplant*. 1998;13:1499–1505.
64. Foley RN, Curtis BM, Randell EW, et al. Left ventricular hypertrophy in new hemodialysis patients without symptomatic cardiac disease. *Clin J Am Soc Nephrol*. 2010;5:805–813.
65. Bansal N, Keane M, Delafontaine P, et al. A longitudinal study of left ventricular function and structure from CKD to ESRD: the CRIC study. *Clin J Am Soc Nephrol*. 2013;8:355–362.
66. Krishnasamy R, Isbel NM, Hawley CM, et al. The association between left ventricular global longitudinal strain, renal impairment and all-cause mortality. *Nephrol Dial Transplant*. 2014;29:1218–1225.
67. Masson S, Latini R, Milani V, et al. Prevalence and prognostic value of elevated urinary albumin excretion in patients with chronic heart failure: data from the GISSI-heart failure trial. *Circ Heart Fail*. 2010;3:65–72.
68. Jackson CE, Solomon SD, Gerstein HC, et al. Albuminuria in chronic heart failure: prevalence and prognostic importance. *Lancet*. 2009;374:543–550.
69. Jackson CE, MacDonald MR, Petrie MC, et al. Associations of albuminuria in patients with chronic heart failure: findings in the Aliskiren Observation of Heart Failure Treatment study. *Eur J Heart Fail*. 2011;13:746–754.
70. Katz DH, Burns JA, Aguilar FG, et al. Albuminuria is independently associated with cardiac remodeling, abnormal right and left ventricular function, and worse outcomes in heart failure with preserved ejection fraction. *JACC Heart Fail*. 2014;2:586–596.
71. Chen Y, Cai L, Du Z, et al. Dipstick proteinuria is a prognostic indicator of short-term mortality in patients with heart failure. *Int J Cardiol*. 2017;230:59–63.
72. Ferreira JP, Santos M, Almeida S, et al. The role of albuminuria as a non-invasive marker for congestive acutely decompensated chronic heart failure and the spironolactone effect in elderly Portuguese: a non-randomized trial. *Nephrology (Carlton)*. 2014;19:149–156.
73. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371:993–1004.
74. Damman K, Gori M, Claggett B, et al. Renal effects and associated outcomes during angiotensin-neprilysin inhibition in heart failure. *JACC Heart Fail*. 2018;6:489–498.
75. Scirica BM, Mosenzon O, Bhatt DL, et al. Cardiovascular outcomes according to urinary albumin and kidney disease in patients with type 2 diabetes at high cardiovascular risk: observations from the SAVOR-TIMI 53 trial. *JAMA Cardiol*. 2017.
76. Damman K, Masson S, Hillege HL, et al. Tubular damage and worsening renal function in chronic heart failure. *JACC Heart Fail*. 2013;1:417–424.
77. Labban B, Arora N, Restaino S, et al. The role of kidney biopsy in heart transplant candidates with kidney disease. *Transplantation*. 2010;89:887–893.
78. Alvelos M, Lourenco P, Dias C, et al. Prognostic value of neutrophil gelatinase-associated lipocalin in acute heart failure. *Int J Cardiol*. 2013;165:51–55.
79. Palazzuoli A, Ruocco G, Beltrami M, et al. Admission plasma neutrophil gelatinase associated lipocalin (NGAL) predicts worsening renal function during hospitalization and post discharge outcome in patients with acute heart failure. *Acute Card Care*. 2014;16:93–101.
80. Park M, Vittinghoff E, Liu KD, et al. Urine biomarkers neutrophil gelatinase-associated lipocalin (NGAL) and kidney injury molecule-1 (KIM-1) have different patterns in heart failure exacerbation. *Biomark Insights*. 2013;8:15–18.
81. Jungbauer CG, Birner C, Jung B, et al. Kidney injury molecule-1 and N-acetyl-beta-D-glucosaminidase in chronic heart failure: possible biomarkers of cardiorenal syndrome. *Eur J Heart Fail*. 2011;13:1104–1110.
82. Damman K, Masson S, Hillege HL, et al. Clinical outcome of renal tubular damage in chronic heart failure. *Eur Heart J*. 2011;32:2705–2712.
83. Breidthardt T, Socrates T, Drexler B, et al. Plasma neutrophil gelatinase-associated lipocalin for the prediction of acute kidney injury in acute heart failure. *Crit Care*. 2012;16:R2.
84. van Deursen VM, Damman K, Voors AA, et al. Prognostic value of plasma neutrophil gelatinase-associated lipocalin for mortality in patients with heart failure. *Circ Heart Fail*. 2014;7:35–42.
85. Mishra J, Dent C, Tarabishi R, et al. Neutrophil gelatinase-associated lipocalin (NGAL) as a biomarker for acute renal injury after cardiac surgery. *Lancet*. 2005;365:1231–1238.
86. Ding H, He Y, Li K, et al. Urinary neutrophil gelatinase-associated lipocalin (NGAL) is an early biomarker for renal tubulointerstitial injury in IgA nephropathy. *Clin Immunol*. 2007;123:227–234.
87. Maisel AS, Wettersten N, van Veldhuisen DJ, et al. Neutrophil gelatinase-associated lipocalin for acute kidney injury during acute heart failure hospitalizations: the AKINESIS study. *J Am Coll Cardiol*. 2016;68:1420–1431.
88. Damman K, Valente MAE, van Veldhuisen DJ, et al. Plasma neutrophil gelatinase-associated lipocalin and predicting clinically relevant worsening renal function in acute heart failure. *Int J Mol Sci*. 2017;18:doi:10.3390/ijms18071470.
89. Damman K, Ng Kam Chuen MJ, MacFadyen RJ, et al. Volume status and diuretic therapy in systolic heart failure and the detection of early abnormalities in renal and tubular function. *J Am Coll Cardiol*. 2011;57:2233–2241.
90. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364:797–805.
91. Damman K, Tang WH, Felker GM, et al. Current evidence on treatment of patients with chronic systolic heart failure and renal insufficiency: practical considerations from published data. *J Am Coll Cardiol*. 2014;63:853–871.
92. Hasin T, Topilsky Y, Schirger JA, et al. Changes in renal function after implantation of continuous-flow left ventricular assist devices. *J Am Coll Cardiol*. 2012;59:26–36.
93. Mehra MR, Canter CE, Hannan MM, et al. The 2016 International Society for Heart Lung Transplantation listing criteria for heart transplantation: a 10-year update. *J Heart Lung Transplant*. 2016;35:1–23.

BOARD REVIEW QUESTIONS

1. A 67-year-old male patient with a medical history of diabetic nephropathy (serum creatinine, 1.3 mg/dL) and hypertension presents to the emergency room with acute shortness of breath, tachypnea, peripheral edema, and jugular venous distention on physical examination. Laboratory tests reveal an acute rise in the serum creatinine level to 4.5 mg/dL and markedly elevated N-terminal prohormone of brain natriuretic peptide (NT-proBNP) levels (4000 pg/L; normal value <300 pg/mL), suggestive of acute heart failure. This clinical condition could be regarded as
 - a. Cardiorenal syndrome type 1
 - b. Cardiorenal syndrome type 2
 - c. Cardiorenal syndrome type 3
 - d. Cardiorenal syndrome type 4
 - e. Cardiorenal syndrome type 5

Answer: c

Rationale: Cardiorenal syndrome type 3 is the clinical condition in which acute (or chronic) kidney injury (AKI) leads to acute heart failure. This has been termed “acute renocardiac syndrome.”

2. The (main) pathophysiology of a reduced glomerular filtration rate (GFR) in patients with chronic heart failure (cardiorenal syndrome type 2) is
 - a. Glomerulosclerosis
 - b. Increased renal venous pressure
 - c. Reduced renal perfusion
 - d. a and c
 - e. b and c

Answer: e

Rationale: In the situation of heart failure with reduced ejection fraction (HFREF), cohort, cross-sectional, and longitudinal studies have shown that the main determinant

of a reduced GFR in these patients is a reduction in renal blood flow. However, in the presence of a low renal blood flow, GFR may also be impaired by a high central (and renal) venous pressure. Although glomerulosclerosis is probably present in patients with heart failure as well, the relative contribution to a reduced GFR is (probably) small.

3. A 79-year-old female patient is diagnosed with HFREF as a consequence of diffuse coronary artery disease. According to heart failure guidelines, she is started on angiotensin-converting enzyme (ACE) inhibitor therapy (enalapril 2.5 mg bid, uptitrated to 5 mg bid after 2 weeks). At the next follow-up visit, the patient feels better but describes occasional postural hypotension, and her serum creatinine level has risen by 0.35 mg/dL (to 1.3 mg/dL). Blood pressure is now 90/60 mm Hg. The next step with regard to her ACE inhibitor therapy is
 - a. Uptitrate ACE inhibitor therapy further
 - b. Decrease dose of ACE inhibitor
 - c. Stop ACE inhibitor
 - d. Continue ACE inhibitor at the same dose
 - e. Switch ACE inhibitor to an angiotensin receptor blocker (ARB.)

Answer: d

Rationale: Some increase in the serum creatinine level with ACE inhibitor therapy is expected and can be accepted if the clinical situation of the patient allows this—that is, there is no evidence of worsening complaints or signs of oliguria and/or hyperkalemia. The serum creatinine level should be checked regularly after the initial increase. Because of her low blood pressure and postural hypotension, up-titration of enalapril dose is not recommended at this point.